

巴适声工厂隐私版使用手册 / Bashi Voice Factory Privacy Edition User Guide

Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

License MIT version 0.1.4 python 3.12 embed platform Windows

Version: 0.1.4

A fully offline desktop web app for high-quality text-to-speech and speech-to-text. After the one-time first-launch download, **everything runs on your own machine** — TTS synthesis, audio export, transcription, and storage. No audio data ever leaves your computer.

⚡ Need cloud-quality voices and faster generation? Check out [Bashi Voice Factory Turbo \(巴适声工厂 · 极速版\)](#) — Microsoft Edge TTS, 14 languages, 50,000-character long text. Requires internet.

Author: Alex Li (ncorecpu@gmail.com) **License:** [MIT License](#) **Source code:** <https://github.com/gtree965/bashi-voice-factory-privacy> · <https://gitee.com/gtree965/bashi-voice-factory-privacy> (mirror for mainland China) **Download:** [GitHub Releases](#) · [Gitee](#) (recommended in mainland China)

What's New in v0.1.4

- **Faster first synthesis:** full-path TTS warmup now starts asynchronously when the UI loads, using a throwaway synthesis rather than model loading alone.

The UI shows a neutral warming state; an early request waits for warmup instead of failing.

- **Fixed the PDF user guide:** the bundled manual no longer carries broken internal links that pointed at directories on the build machine. All links in the manual are now public HTTPS addresses.
 - **Build-time guard:** packaging now refuses to ship a user guide containing any non-HTTPS link, so this class of defect cannot recur.
 - **Gitee mirror:** source and downloads are now also served from Gitee for users in mainland China.
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✨ Highlight Features

🔒 Fully Local Text-to-Speech

- **Local Qwen3-TTS-12Hz-1.7B-CustomVoice** running entirely on your machine — no cloud calls during synthesis.
- **9 curated voice presets** with native-language sample previews built in.
- **10 languages** supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
- **Instruction control** for style, accent, emotion via natural-language prompts.
- **Streaming controls for long text and Shadowing Mode:** stop synthesis while keeping completed chunks, auto-play the first ready chunk by default, and continue playback as later chunks arrive; GGUF long text also preserves a complete seekable/downloadable MP3 after the streamed preview.

🎯 Automatic Backend Selection

- **GGUF + Vulkan** for AMD / Intel / iGPU users (default, fast, RAM-friendly).
- **GGUF + CUDA** for NVIDIA users via a one-click in-app upgrade (optional ~595 MB add-on download; no manual weight setup). New in v0.1.1.
- **Reliable NVIDIA detection and startup probes:** v0.1.2 detects the physical NVIDIA GPU through virtual-display layers and isolates native probe crashes so the launcher can report an actionable error instead of disappearing.
- **CPU fallback** for entry-level hardware (works on Intel N100-class boxes).
- One-click **speed test** calibrates ETA estimates to your specific machine.

Local Offline Speech-to-Text (STT)

- **SenseVoice Small** (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).
- **Parakeet TDT 0.6B** (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
- **VAD-based segmentation** via Silero — accurate timestamps, no chunk-boundary stutter.
- **Live transcript streaming** via SSE; exports to TXT / SRT / VTT.

Smart First-Launch Download with Resume

- **HTTP Range / resume** built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.
- **3-attempt retry** with exponential backoff for both pip dependencies and model files.
- **China-friendly mirrors**: pip falls back through Aliyun → Tsinghua → PyPI when a mirror is unreachable or refuses your IP; GGUF runtime hosted on ModelScope; STT models hosted on hf-mirror.cn.

LAN Sharing for Phone / Tablet Use

- First-launch prompt: bind to 0.0.0.0 to allow access from any device on the same WiFi.
 - Compatible with mobile browsers — type the LAN IP and use it like a local web app.
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Quick Start

Easiest path

1. Unzip the downloaded file to any folder (Desktop or wherever you like).
2. **Double-click start_启动.bat** at the top level of the extracted folder.
3. First launch will:
 - Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)
 - Automatically download the GGUF model when missing (~2.2 GB, 2-15 min depending on network; Ctrl+C cancels and the next launch resumes)

- Open `http://127.0.0.1:5050` in your default browser
4. After first launch, **all subsequent launches are offline** and start in ~5 seconds.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

Network Behavior

Privacy commitment: this app does not phone home, does not upload audio, does not collect telemetry. **The app never automatically checks any internet resource.** Every network operation is documented below and requires explicit user action — no background polling, no analytics, no silent update probes.

First launch only (one-time, ~700 MB + ~2.2 GB):

- pip dependencies from `https://mirrors.aliyun.com/pypi/simple/`, falling back to Tsinghua then PyPI (PyPI only, outside China)
- GGUF model from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
- STT model (when user opts in) from `https://hf-mirror.com/csukuangfj/...` (or HuggingFace fallback)

Runtime network activity: zero. TTS synthesis, STT transcription, and audio export are 100% on-device.

Optional CUDA acceleration (NVIDIA users, user-initiated only): when an NVIDIA GPU is detected, the UI shows an “Upgrade to CUDA acceleration” banner. Clicking it downloads a one-time ~595 MB CUDA runtime add-on from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime`. This is entirely optional — the Vulkan path works on NVIDIA without it. Nothing downloads unless you click.

Manual update check (user-initiated only, never automatic): the UI footer “Check for updates” button opens a release page in a new browser tab. You can also check manually anytime at:

- GitHub Releases: <https://github.com/gtree965/bashi-voice-factory-privacy/releases>
- Gitee: <https://gitee.com/gtree965/>

Air-gapped / sensitive deployment (banks, healthcare, government, internal corporate networks): after first-launch downloads complete on a connected machine, the entire bashi-voice-factory-privacy-v0.x.0/ extracted folder can be copied to an offline / physically-isolated machine and run with zero further network access required. The app has no telemetry endpoints to firewall, no auto-update worker, no callback URLs. To verify, monitor outbound traffic with Wireshark / Resource Monitor during a normal session — it should be silent.

Hardware Expectations

The table below shows **actually measured** numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	“你好。” probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	✓ Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	✓ Tested
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	✓ Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	✓ Tested 2026-05-25
NVIDIA RTX 5070 (12 GB, cloud PC, Blackwell)	GGUF + Vulkan/CUDA + DirectML	~1 s	41 s - 1 m 17 s	✓ Tested 2026-06-12 (Vulkan) & 2026-07-24 (海马云 HMv

Hardware	Auto-selected backend	“你好。” probe (25 chars)	1,000-char synthesis ETA	Status
				Cloud PC): full cold-start synthesis confirmed; optional CUDA add-on verified working (LLM 28 layers on CUDA0, decoder DirectML fp16). CUDA-vs-Vulkan A/B still welcome.
Other NVIDIA RTX / GTX (desktop)	GGUF + Vulkan + DirectML · optional CUDA add-on	community reports welcome	community reports welcome	awaiting desktop tester reports via GitHub Issues
Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated yet

⚠ **Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation.** A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).

i NVIDIA users (desktop RTX / GTX): the default path is GGUF + Vulkan (NVIDIA cards support Vulkan via the proprietary driver). For native CUDA acceleration, click the one-click in-app upgrade banner (v0.1.1+) — it downloads a ~595 MB CUDA runtime add-on, no manual weight setup required. **Requires NVIDIA driver ≥ 545.x for the CUDA 12.4 runtime.** Vulkan path measured on RTX 5070 (cloud PC): **1 s for the 25-char probe**, so the default Vulkan experience is already excellent on a desktop NVIDIA

card. CUDA add-on A/B numbers + reports from other RTX/GTX cards welcome via [GitHub Issues](#). Cloud datacenter NVIDIA cards (A10/A100/T4) running in TCC mode have additional setup steps — see Troubleshooting below.

🧩 Hardware Coverage Matrix

Beyond the specific machines benchmarked above, here is how the auto-selected acceleration path maps across mainstream Windows hardware:

Hardware class	Acceleration path used	Status
NVIDIA RTX 30 / 40 / 50 (desktop)	GGUF + Vulkan default · optional CUDA in-app upgrade	✅ RTX 5070 (Blackwell) verified on 海马云 cloud PC — 2026-06-12 Vulkan probe ~1 s; 2026-07-24 full cold-start synthesis + optional CUDA add-on confirmed working (LLM 28 layers on CUDA0, decoder DirectML fp16, driver 610.47). CUDA-vs-Vulkan A/B numbers + other-card reports still welcome via Issues. CUDA add-on requires driver ≥ 545.x.
NVIDIA GTX 10 / 16 (desktop)	GGUF + Vulkan default · optional CUDA in-app upgrade	✅ Same flow, same driver requirement; reports welcome via Issues.
NVIDIA datacenter (A10 / A100 / T4)	Manual setup required	⚠️ TCC mode + old cloud driver = manual workaround. See Troubleshooting.
AMD RX 500 / 600 / 7000 / 9000 (discrete)	GGUF + Vulkan + DirectML	✅ Tested (RX 590, RX 9060 XT)

Hardware class	Acceleration path used	Status
Intel Arc A-series (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
Intel iGPU (UHD / Iris Xe / Arc iGPU)	GGUF + Vulkan + DirectML	✅ Tested (Intel N305 UHD)
AMD APU iGPU (Vega 7/8, RDNA 2/3)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
CPU only (no usable GPU / no driver)	GGUF + CPU (ggml-cpu auto-SIMD)	✅ Works; auto-selected only when no GPU detected
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	❌ Not yet utilized
ARM64 Windows (Snapdragon X Copilot+ PC)	—	❌ Not supported (runs via slow x64 emulation)

⚙️ Known Limitations (roadmap)

- **NVIDIA CUDA is opt-in, not bundled.** By default NVIDIA cards use the Vulkan path; native CUDA acceleration requires the one-click in-app upgrade (~595 MB add-on, v0.1.1+). This keeps the main download small for the AMD / Intel / CPU majority. Requires NVIDIA driver ≥ 545.x.
 - **Weak iGPU may be slower than pure CPU.** On Intel N100-class hardware, DirectML / Vulkan transfer + scheduling overhead can outweigh GPU benefit. A/B test by launching from a cmd window after `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` — feedback on real speed numbers welcome.
 - **NPU acceleration is not yet used** (Snapdragon X Elite, Intel Lunar Lake AI Boost, AMD Ryzen AI 300). Under investigation for v0.2+.
 - **ARM64 Windows is not natively supported** (Surface Pro 11, Galaxy Book4 Edge, etc.). x64 emulation works but is slow. Native ARM64 build is a v0.3+ candidate.
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Troubleshooting

Symptom	Likely cause / fix
Start_启动.bat shows "Array index expression is missing"	Old zip — re-download the latest from GitHub Releases or Gitee; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	Retry triggers automatically (3 attempts, 5s/30s/120s backoff); all 3 fail, fix network and re-run the launcher — pip caches what's already installed
pip install fails with a "long path support" message	Run the registry one-liner from the launcher's advisory in an Administrator PowerShell, then re-launch
pip install fails on every mirror, or a mirror refuses your IP (403 / denied by IP ACL)	The launcher already retries Aliyun → Tsinghua → PyPI. To force one index, set BASHI_PIP_INDEX_URL before launching, e.g. set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/simple
GGUF download interrupted	Same: retries with HTTP Range/resume; re-run launcher to continue from .part file
App exits with "No usable backend was found"	Check app.log (structured, timestamped application logs) and launch_log.txt (launcher steps and raw stderr) — usually GGML runtime DLL missing, GPU driver outdated, or RAM <8 GB. The app now prints a bilingual advisory with specific causes
STT download says "镜像失败"	Should not happen in v0.1.0 final — old behavior from remote ModelScope path
Cloud / datacenter NVIDIA GPU (A10 / A100 / T4 on Chinese cloud Windows images): access violation reading 0x0000000000000000 Or GGUF probe failed	Datacenter NVIDIA GPUs typically run in TCC mode (Vulkan/DirectML disabled) with older drivers (~538.x) incompatible with the CUDA 12.4 add-on. Desktop RTX/GTX cards are NOT affected (they run WDDM mode by default with modern drivers). In v0.1.2, native startup-probe crashes are isolated and reported instead of terminating the launcher, but TCC setup still requires the manual workaround: (1) rename vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll to .disabled; (2) edit bashi-privacy-app\run_portable.ps1 and

Symptom	Likely cause / fix
	<code>\$env:USE_GGUF_BACKEND = "1" + \$env:GGUF_ONNX_PROVIDER =</code> after the <code>Remove-Item Env:USE_GGUF_BACKEND</code> lines (~line 270) pre-install the CUDA add-on via CLI python <code>download_cuda_runtime.py</code> . CUDA 12.4 requires NVIDIA drive 545.x. Automatic TCC handling remains planned for a later patch.

Full log paths: bashi-privacy-app\app.log and bashi-privacy-app\launch_log.txt

What's in the Zip

bashi-voice-factory-privacy-v0.1.4/	
├─ Start_启动.bat	← double-click here
├─ Start_CPU_only_仅CPU启动.bat	← force CPU mode
(entry-level iGPU A/B)	
├─ README.md	← this file
├─ README_CN.md	← Chinese version
├─ LICENSE	← MIT license
├─ VERSION	← release version
├─ 巴适声工厂隐私版使用手册	
_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf	
	← help PDF
(bilingual)	
├─ bashi-privacy-app/	← app code +
embedded Python	
└─ run_portable.bat	← same launcher,
direct path	
└─ ...	
└─ vulkan_backend_spike/	← GGUF runtime
(populated on first launch)	
└─ Qwen3-TTS-GGUF/	

License

MIT License © 2026 Alex Li

Bundled third-party components retain their original licenses:

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**: Tongyi Lab (Alibaba), Apache 2.0
 - **GGUF runtime**: based on HaujetZhao/Qwen3-TTS-GGUF
 - **SenseVoice Small / Silero VAD / Parakeet TDT**: see respective model cards
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Issues, feedback, and feature requests welcome via the [GitHub release page](#) or email.

巴适声工厂 · 隐私版 (Bashi Voice Factory Privacy Edition)

License MIT version 0.1.4 python 3.12 embed platform Windows

版本: 0.1.4

完全离线运行的本地语音工厂网页应用。首次启动联网下载完依赖与模型之后，**所有文字转语音、语音转文字、音频生成、文件保存都在你自己的电脑上完成**，没有任何音频数据上传云端。

⚡ 想要云端语音质量、更快的合成速度？请关注 [巴适声工厂 · 极速版 \(Bashi Voice Factory Turbo\)](#) — 微软 Edge TTS 引擎、14 种语言、5 万字长文。需要联网。

作者: Alex Li (ncorecpu@gmail.com) **许可:** [MIT License](#) **源码:**

<https://github.com/gtree965/bashi-voice-factory-privacy> ·

<https://gitee.com/gtree965/bashi-voice-factory-privacy> (国内镜像) **下载:**

[GitHub Releases](#) · [Gitee](#) (国内推荐)

v0.1.4 更新内容

- **首次合成更快:** 界面加载时即异步预热完整 TTS 链路，用一次丢弃式合成代替单纯加载模型。界面显示中性的预热状态；预热期间的早到请求会等待，而不是直接失败。
 - **修复 PDF 使用手册:** 随包分发的手册不再带有指向构建机器本地目录的失效内链，手册内所有链接均为公开 HTTPS 地址。
 - **构建期门禁:** 打包时若手册中存在任何非 HTTPS 链接即拒绝出包，杜绝此类缺陷复发。
 - **Gitee 镜像:** 源码与下载新增 Gitee 入口，方便中国大陆用户访问。
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🌟 核心亮点

🔒 完全本地文字转语音

- **Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理** — 合成过程零云端调用。
- **9 个精选音色**，内置母语试听样例。
- **10 种语言**：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
- **自然语言指令控制** — 风格、口音、情绪都可通过 instruct 提示调整。
- **长文与跟读流式控制** — 可停止合成并保留已完成片段；默认自动播放首个就绪片段，后续片段生成后自动续播；GGUF 长文试听结束后仍保留可拖动、可下载的完整 MP3。

🎯 后端自动选择

- **GGUF + Vulkan**：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
- **GGUF + CUDA**：NVIDIA 用户一键应用内升级（可选 ~595 MB 附加包下载，无需手动配置权重）。v0.1.1 新增。
- **更可靠的 NVIDIA 检测与启动探测**：v0.1.2 可穿透虚拟显示层识别真实 NVIDIA 显卡，并隔离原生探测崩溃，让启动器显示可处理的错误而不是直接消失。
- **CPU 回退**：入门级硬件（如 Intel N100 系列）也能跑。
- 一键 **测速** 校准 ETA 估算，让等待时间贴合你这台机器。

🎧 本地离线音视频转文字 (STT)

- **SenseVoice Small**（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
- **Parakeet TDT 0.6B**（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
- **Silero VAD 精准分段** — 时间戳准确、零重叠、零卡顿。
- **实时滚屏字幕 SSE 流式输出**；支持导出 TXT、SRT、VTT。

💡 智能首次下载，断点续传

- **HTTP Range / resume 内置** — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。
- **3 次重试，指数退避** — pip 依赖与模型文件都有同样的兜底逻辑。

- **国内友好镜像**：pip 按阿里云 → 清华 → PyPI 依次回退，某个镜像不可达或拒绝本机 IP 时自动换源；GGUF 运行模型走 ModelScope；STT 模型走 hf-mirror.cn。

局域网共享，手机平板可访问

- 首次启动询问是否绑定 0.0.0.0，允许同一 WiFi 下其他设备访问。
 - 手机浏览器输入电脑的局域网 IP 即可使用，跟本地网页一样流畅。
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快速上手

最简流程

1. 把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。
2. 在解压后的顶层目录里 **双击** Start_启动.bat。
3. 首次启动会：
 - 安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）
 - 缺少 GGUF 模型时自动下载（约 2.2 GB，2-15 分钟；可按 Ctrl+C 取消，下次启动会续传）
 - 自动在默认浏览器打开 <http://127.0.0.1:5050>
4. 首次完成后，**之后每次启动都是离线**，5 秒左右即可打开。

进阶

如果想跳过顶层启动器，可以直接运行 bashi-privacy-app\run_portable.bat，效果完全一样。

网络行为说明

隐私承诺：本程序不主动联网、不上传音频、不收集使用数据。**程序永远不会自动检查任何在线资源**。所有联网行为均在下文有完整说明，且需要用户主动触发——无后台轮询、无统计分析、无静默更新探测。

仅首次启动（一次性，约 700 MB + 约 2.2 GB）：

- pip 依赖来自 <https://mirrors.aliyun.com/pypi/simple/>，失败时依次回退清华、PyPI（海外用户仅用 PyPI）

- GGUF 模型来自 <https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime>
- STT 模型（用户主动点击下载）来自 <https://hf-mirror.com/csukuangfj/...>（备用 HuggingFace）

运行时网络行为：零。文字转语音、音视频转文字、音频导出全部本机完成。

可选 CUDA 加速（NVIDIA 用户，仅用户主动触发）：检测到 NVIDIA 显卡时，UI 会显示“升级到 CUDA 加速”横幅。点击后从

<https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime> 下载一次性 ~595 MB CUDA 运行时附加包。完全可选 — 不下载也能用 Vulkan 路线。不点就不会下载任何东西。

手动更新检查（仅用户主动触发，永不自动）：UI 右下角“查看最新版本”按钮在新浏览器标签页打开发布页。也可以随时手动访问：

- GitHub Releases: <https://github.com/gtree965/bashi-voice-factory-privacy/releases>
- Gitee: <https://gitee.com/gtree965/>

离网 / 敏感场景部署（银行、医疗、政务、企业内网）：在联网机器上完成首次启动的所有下载后，整个 bashi-voice-factory-privacy-v0.x.0/ 解压文件夹可以原样拷贝到离网/物理隔离机器上运行，之后所有功能均无需任何网络访问。程序没有任何遥测端点需要防火墙拦截，没有自动更新守护进程，也没有回调 URL。可用 Wireshark / 资源监视器在正常使用过程中监测出站流量验证 — 应当全程静默。

硬件要求

下表是**作者实测**的数据。首次运行后点右侧面板的“测速”按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	“你好。”探测 (25 字)	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	✓ 2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	✓ 实测

硬件	自动选择的后端	“你好。” 探测 (25 字)	1,000 字合成预计	状态
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	✓ 2026-05-25 实测
Intel N100 迷你主机 + UHD 集显	GGUF + Vulkan / DirectML	126 秒	58 分钟 - 1 小时 49 分	✓ 2026-05-25 实测
NVIDIA RTX 5070 (12 GB, 云电脑, Blackwell)	GGUF + Vulkan/CUDA + DirectML	约 1 秒	41 秒 - 1 分 17 秒	✓ 2026-06-12 (Vulkan) & 2026-07-24 (海马云 HMv Cloud PC) 实测：全冷启动合成确认；可选 CUDA 附加包已实证可用 (LLM 28 层跑 CUDA0、解码 DirectML fp16) 。CUDA-vs-Vulkan A/B 仍欢迎反馈。
其它 NVIDIA RTX / GTX (桌面)	GGUF + Vulkan + DirectML · 可选 CUDA 附加包	待社区报数	待社区报数	等待桌面 tester 通过 GitHub Issues 报数
Apple Silicon / Intel Arc	—	暂未实测	暂未实测	尚未验证

⚠️ **入门级 CPU (Intel N100 / N305 一类) 仅适合短句试用 — 单次生成建议不超过 200 字。** 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频 (讲座、有声书)，请用独立显卡硬件 (AMD RX 500/600/9000

系、NVIDIA RTX 类) 。

i NVIDIA 用户 (桌面 RTX / GTX) : 默认路线是 GGUF + Vulkan (NVIDIA 驱动支持 Vulkan) 。想要原生 CUDA 加速, 点击应用内一键升级横幅 (v0.1.1+) 一会下载 ~595 MB CUDA 运行时附加包, 无需手动配置权重。**需要 NVIDIA 驱动 $\geq 545.x$ 才能使用 CUDA 12.4 运行时。** Vulkan 路径在 RTX 5070 (云电脑) 实测: **25 字探测 1 秒**, 桌面 NVIDIA 用户的默认 Vulkan 体验已经非常顺畅。CUDA 附加包 A/B 数字 + 其它 RTX/GTX 卡的报数欢迎通过 [GitHub Issues](#) 反馈。云端数据中心 NVIDIA 显卡 (A10/A100/T4) 在 TCC 模式下需要额外设置 — 见下方常见问题排查。

🔗 硬件兼容性矩阵

除了上表里具体测过的几台机器, 下表说明主流 Windows 硬件在自动后端选择下走的加速路径:

硬件类别	实际加速路径	状态
NVIDIA RTX 30 / 40 / 50 (桌面)	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	✅ RTX 5070 (Blackwell) 海马云云电脑已验证——2026-06-12 Vulkan 探测约 1 秒; 2026-07-24 全冷启动合成 + 可选 CUDA 附加包实证可用 (LLM 28 层跑 CUDA0、解码 DirectML fp16、驱动 610.47) 。CUDA-vs-Vulkan A/B 数字 + 其它卡报数仍欢迎通过 Issues 反馈。CUDA add-on 需驱动 $\geq 545.x$ 。
NVIDIA GTX 10 / 16 (桌面)	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	✅ 同一流程, 同样驱动要求; 报数欢迎通过 Issues。
NVIDIA 数据中心卡 (A10 / A100 / T4)	需要手动配置	⚠️ TCC 模式 + 云端老驱动 = 需要手动 workaround。见常见问题排查。

硬件类别	实际加速路径	状态
AMD RX 500 / 600 / 7000 / 9000 (独立显卡)	GGUF + Vulkan + DirectML	✅ 实测 (RX 590、RX 9060 XT)
Intel Arc A 系列 (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
Intel 集显 (UHD / Iris Xe / Arc iGPU)	GGUF + Vulkan + DirectML	✅ 实测 (Intel N305 UHD)
AMD APU 集显 (Vega 7/8、RDNA 2/3)	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
纯 CPU (无可用 GPU / 无驱动)	GGUF + CPU (ggml-cpu 自动 SIMD)	✅ 可用; 仅在无 GPU 时自动选择
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	❌ 暂未利用
ARM64 Windows (Snapdragon X Copilot+ PC)	—	❌ 暂不支持 (走 x64 仿真较慢)

⚙️ 已知限制 (路线图)

- **NVIDIA CUDA 为可选项，不内置。** NVIDIA 显卡默认走 Vulkan 路线; 原生 CUDA 加速需点击应用内一键升级 (~595 MB 附加包, v0.1.1+)。这样主下载包对 AMD / Intel / CPU 大多数用户保持精简。需 NVIDIA 驱动 ≥ 545.x。
 - **弱集显可能比纯 CPU 还慢。** Intel N100 类硬件上, DirectML / Vulkan 的数据传输与调度开销可能盖过 GPU 收益。可在 cmd 窗口里 set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU 后再启动来 A/B 测试, 欢迎反馈实测数字。
 - **NPU 加速尚未利用** (Snapdragon X Elite、Intel Lunar Lake AI Boost、AMD Ryzen AI 300)。v0.2+ 调研中。
 - **ARM64 Windows 暂未原生支持** (Surface Pro 11、Galaxy Book4 Edge 等)。x64 仿真能跑但速度受限。原生 ARM64 构建是 v0.3+ 候选。
-

常见问题排查

现象	可能原因 / 处理
双击 Start_启动.bat 报 “Array index expression is missing”	旧版 zip。请从 GitHub Releases 或 Gitee 重新下载最新版；BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	自动重试 3 次（5/30/120 秒退避）。3 次都失败时，修好网重新运行启动器即可（pip 会跳过已装好的包）
pip 安装失败并提示 “long path support”	按启动器提示，在管理员 PowerShell 中运行一次注册表命令后重新启动
所有镜像都装不上，或镜像拒绝本机 IP（403 / denied by IP ACL）	启动器已按阿里云 → 清华 → PyPI 自动换源。若要指定某一源，启动前设置 BASHI_PIP_INDEX_URL，例如 set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/si
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器 .part 文件续传
启动报 “No usable backend was found”	查 app.log（带时间戳的结构化应用日志）和 launch_log.txt（启动器步骤与原生 stderr）。常见原因：GGUF 运行 DLL 缺失、卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告
STT 下载闪过 “镜像失败”	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，除
云端 / 数据中心 NVIDIA 显卡（A10 / A100 / T4，国内云 Windows 镜像）：access violation reading 0x0000000000000000 或 GGUF probe failed	数据中心 NVIDIA 显卡通常运行在 TCC 模式 （Vulkan/Direct 被屏蔽），且驱动版本较旧（~538.x）跟 CUDA 12.4 add-on 兼容。 桌面 RTX/GTX 显卡不受影响 （默认 WDDM 模式 + 3 驱动）。v0.1.2 会隔离并报告原生启动探测崩溃，不再让启动器直接退出，但 TCC 配置仍需手动 workaround：(1) 把 vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll 重命名 .disabled；(2) 编辑 bashi-privacy-app\run_portable.ps1，在 Remove-Item Env:USE_GGUF_BACKEND 行（约第 270 行）之后加 \$env:USE_GGUF_BACKEND = "1" + \$env:GGUF_ONNX_PROVIDER = "CPU"；(3) 命令行预装 CUDA add-on：python

现象	可能原因 / 处理
	download_cuda_runtime.py。CUDA 12.4 需要 NVIDIA 驱动 $\geq 545.x$ 。TCC 自动处理仍计划在后续补丁实现。

完整日志路径：bashi-privacy-app\app.log 和 bashi-privacy-app\launch_log.txt

zip 包目录结构

bashi-voice-factory-privacy-v0.1.4/	
├─ Start_启动.bat	← 双击这里
├─ Start_CPU_only_仅CPU启动.bat	← 强制 CPU 模式（入门集显 A/B 对比用）
├─ README.md	← 英文文档
├─ README_CN.md	← 本文件
├─ LICENSE	← MIT 许可证
├─ VERSION	← 发布版本号
├─ 巴适声工厂隐私版使用手册	
_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf	
	← 中英双语帮助 PDF
├─ bashi-privacy-app/	← 程序代码 + 嵌入式
Python	
└─ run_portable.bat	← 等效启动器（直接路径）
└─ ...	
└─ vulkan_backend_spike/	← GGUF 运行时（首次启动自动填充）
└─ Qwen3-TTS-GGUF/	

许可

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第三方组件保留各自原始许可：

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**：通义实验室（阿里巴巴）Apache 2.0
- **GGUF runtime**：基于 HaujetZhao/Qwen3-TTS-GGUF

- **SenseVoice Small / Silero VAD / Parakeet TDT**: 详见各自模型卡
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欢迎通过 [GitHub release 页面](#)或邮件提交 issue、反馈或功能建议。